REPORT

Student Details

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Course Code: CSA1717

Course Name: Artificial Intelligence

Assessment: Tool 1 – Analytical Problem Solving

Introduction

This assessment focuses on applying the basic concepts of Artificial Intelligence to solve real-world analytical problems. Different AI techniques such as intelligent agents, search algorithms, problem formulation, and path-finding methods were studied and implemented using Python. The objective was to understand how AI solves problems systematically by selecting suitable algorithms and agents.

Objectives

- To understand AI problem-solving techniques.
- To apply intelligent agent concepts to real-world applications.
- To implement search algorithms using Python.
- To formulate AI problems and identify suitable solution methods.
- To analyze and interpret the obtained results.

Problems Covered

Problem 1 – Water Jug Problem

The Water Jug Problem was solved using a Simple Problem-Solving Agent. A sequence of valid operations was performed to obtain exactly 2 gallons of water in the 4-gallon jug.

Problem 2 – Mars Rover Intelligent Agent

The Mars Rover was analyzed as an intelligent agent. Its percepts, environment, actions, performance measures, and Goal-Based Agent architecture were identified and explained.

Problem 3 – 8-Queens Problem

The classical 8-Queens Problem was solved using the Backtracking Search strategy. The algorithm successfully placed eight queens on the chessboard without conflicts.

Problem 4 – OLA Cab Booking

An OLA Cab Booking system was modeled using a Goal-Based Agent. The booking process, driver selection, and pseudocode were developed to demonstrate intelligent decision-making.

Problem 5 – Uniform Cost Search (UCS)

Uniform Cost Search was implemented to find the least-cost path in a logistics delivery network. The algorithm selected the optimal route based on the minimum cumulative cost.

Software Requirements

- Python 3.x
- Python IDLE / VS Code
- Windows 10/11

Algorithms Used

Problem	Algorithm / AI Technique
Water Jug Problem	Simple Problem-Solving Agent
Mars Rover	Goal-Based Intelligent Agent
8-Queens Problem	Backtracking Search
OLA Cab Booking	Goal-Based Agent
Logistics Network	Uniform Cost Search (UCS)

Aplications

The Artificial Intelligence concepts used in this assessment have many real-world applications:

1. **Water Jug Problem**
   - Used to understand AI problem-solving and state-space search.
   - Applied in planning and decision-making systems.
2. **Mars Rover Intelligent Agent**
   - Used in space exploration and autonomous robots.
   - Helps in scientific research and remote environment exploration.
3. **8-Queens Problem**
   - Used in Constraint Satisfaction Problems (CSP).
   - Applied in scheduling, resource allocation, and optimization problems.
4. **OLA Cab Booking**
   - Used in intelligent transportation systems.
   - Helps in ride allocation, route planning, and customer service applications.
5. **Uniform Cost Search (UCS)**
   - Used in GPS navigation and route optimization.

    - Applied in logistics, transportation, robotics, and network routing.

Advantages

The AI techniques used in this assessment provide several advantages:

- Finds optimal and efficient solutions to complex problems.
- Reduces human effort by automating decision-making.
- Improves accuracy and minimizes errors.
- Helps solve real-world planning and optimization problems.
- Enables intelligent systems to work autonomously.
- Provides systematic and logical problem-solving approaches.
- Supports applications in robotics, transportation, healthcare, and logistics.
- Enhances the efficiency of search algorithms and intelligent agents.

Learning Outcomes

After completing this assessment, the following concepts were understood:

- Intelligent Agents
- Problem Formulation

- State Space Search
- Backtracking Algorithm
- Uniform Cost Search
- Goal-Based Agent
- AI Problem Solving
- Python Implementation of AI Concepts

Results

All five problems were successfully analyzed and solved using appropriate Artificial Intelligence techniques. The Python programs produced the expected outputs, and the selected algorithms successfully achieved their respective objectives.

Conclusion

This assessment provided practical knowledge of Artificial Intelligence fundamentals, including intelligent agents and search techniques. By implementing the Water Jug Problem, Mars Rover analysis, 8-Queens Problem, OLA Cab Booking system, and Uniform Cost Search algorithm, the objectives of the assessment were successfully achieved. The work strengthened the understanding of AI problem-solving methods and their real-world applications.